Random walks

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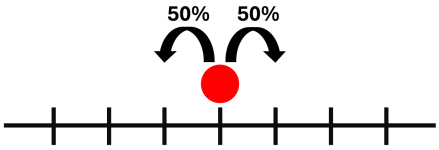
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1 Introducing Random Walks

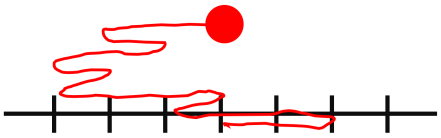
Today we will learn about random walks. Random walks are a powerful mathematical/computational tool with many applications. In biophysics, they can be used to model polymers like DNA and describe the motion of particles, including cells or bacteria. For today we will just focus on building an intuition for random walks and doing some of the math.

1.1 What is a random walk?

A random walk is essentially a series of steps in which the direction of each step is chosen randomly. So what does this look like in 1 dimension?



(a) A random walker choosing which way to go



(b) An example of one random walk path

Figure 1: Random walk in 1 dimension

Well, in one dimension, a random walk is quite simple. For every step of time, a walker decides whether to take a step to the right or to the left, with a fixed step size. For a system like this, there are only 2 important arguments: The step size and the number of steps.

Table 1: A Simple Example Table

Argument	Math	Polymer	Moving particle
Number of steps	N	Length of polymer	Time the particle moves
Step size	a	Persistence length	Characteristic Δx

1.2 Using random walks

When we use random walks, we generally average over many individual results to get measurements. This is done because it is not a good idea to trust the result of just a few randomly generated numbers as the truth. So, to get measurements that can be trusted, we look at statistics, and take averages of many random walks to find results. In notation, we represent this as something like $< X >$, where the <> symbol means "the average of whatever is inside". Given that each random walk has some X that we can measure, then we can write

$$< X >= \frac{1}{M} \sum_{i=1}^{M} X_i$$

, where X_i is the measured X for the random walk i, and we have M random walks.

So what do we actually end up measuring? (What is X in the above equation?) For a 1-d random walk, we generally want to look at the average distance that random walks will go. So let's look at the variable R, which we will define as such:

$$R = x_0 - x_N$$

Here $x_0, x_1, x_2, ..., x_N$ are the positions of the random walker at each step it takes. With this knowledge, let's take a look at the quantity $< R >$. To recap all the previous jargon, this is the average displacement from the starting point to the ending point for many random walks. Below is the mathematical notation of this, with x_{ij} representing the position of the ith random walk at the jth step.

$$< R >= \frac{1}{M} \sum_{i=1}^{M} R_i = \frac{1}{M} \sum_{i=1}^{M} x_{i0} - x_{iN}$$

We can simplify this further by setting x_0 to 0 for every random walk, meaning each random walk will start at 0. This means we can write $< R >$ as such:

$$< R >= \frac{1}{M} \sum_{i=1}^{M} x_{iN}$$

Questions:

1. Do you have any expectations for what the value of $< R >$ should be as M approaches ∞?

2. If not, think about the average displacement after only one step. Keep in mind that displacement can be negative.

3. Is $< R >$ a useful thing to measure to find the average distance traveled? If so, why? If not, what else could we measure to get a better idea?

If you are looking at this page before answering the questions, there are spoilers ahead. Basically, since each step is random forward and back, the end points of the walk will be randomly distributed around the starting point, and so the average R, or $< R >$, will just be the same as the starting point, so $< R >= 0$. Looking at $< R >$ is looking at the average displacement, where really we want to look more at the average distance.

So, what can we look at to get a better idea of the average distance? Well, your first thought may be to just take the absolute value of R, so that the negatives don't end up putting everything at 0, and calculating $< |R| >$. This is a good first thought, but it turns out that our math will be easier if we instead look at $\sqrt{< R^2 >}$.

So, let's try to calculate $< R^2 >$. It will be useful in this case to write use

$$x_{iN} = \Delta x_{i1} + \Delta x_{i2} + ... + \Delta x_{iN}$$

where each Δx_{ij} represents the jth step of the ith random walk. This means each Δx is either a or $-a$, with a 50% chance of each. Feel free to ask me questions, but this derivation should be done by you, using the below as your starting point:

$$< R^2 >= \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} R^2 = \lim_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} {x_{iN}}^2$$

Next, let's repeat this for 2 dimensions. In 2 dimensions, we have a random walk that goes in both x and y. What this means is that for every step, the walker chooses between up, down, left, and right. This means each given step will have a coordinate pair x_{ij}, y_{ij} where one will be 0 and the other $\pm a$. So, $R_i = \sqrt{x_{iN}^2 + y_{iN}^2}$. Now, this may seem like it will change a lot, but the math will be quite similar. Go ahead and calculate $\sqrt{< R^2 >}$ for a random walk in 2 dimensions.

Okay, now that that is done, we will just extrapolate this result for any number of dimensions. And this is true, the answer for $\sqrt{< R^2 >}$ is the same in all dimensions.

Questions:

1. What are $< R >$ and $\sqrt{< R^2 >}$ for 3 dimensional random walks?
2. If you haven't done any coding yet for this project, can you think of a way to test this equation?
3. Write a summary of random walks